

Dark States in Quantum Systems

1 Introduction

In quantum physics, a **dark state** is a coherent superposition of quantum states that does not interact with (or absorb) electromagnetic radiation. These states arise due to destructive interference between different transition pathways, effectively making the system “invisible” to certain external fields.

Dark states play an important role in quantum optics, atomic physics, and quantum information processing.

2 Basic Concept

Consider a three-level atomic system in a Λ configuration with states:

$$|1\rangle, \quad |2\rangle, \quad |3\rangle$$

where:

- $|1\rangle$ and $|2\rangle$ are lower-energy (ground) states,
- $|3\rangle$ is an excited state.

Two laser fields couple the transitions:

$$|1\rangle \leftrightarrow |3\rangle, \quad |2\rangle \leftrightarrow |3\rangle$$

3 Definition of Dark State

A dark state $|D\rangle$ is a superposition of $|1\rangle$ and $|2\rangle$ such that the transition amplitude to $|3\rangle$ cancels out.

It can be written as:

$$|D\rangle = \frac{\Omega_2 |1\rangle - \Omega_1 |2\rangle}{\sqrt{|\Omega_1|^2 + |\Omega_2|^2}}$$

where:

- Ω_1 and Ω_2 are Rabi frequencies of the two laser fields.

4 Key Property

The defining feature of a dark state is:

$$H_{\text{int}} |D\rangle = 0$$

meaning it does not couple to the excited state $|3\rangle$ under the interaction Hamiltonian.

As a result:

- No absorption occurs,
- The system remains trapped in this state.

5 Physical Interpretation

The system has two possible pathways to reach the excited state. In a dark state:

- The probability amplitudes for these pathways interfere destructively,
- The excitation is completely suppressed.

6 Applications

Dark states are central to several important phenomena:

6.1 Coherent Population Trapping (CPT)

Population gets trapped in the dark state, preventing excitation.

6.2 Electromagnetically Induced Transparency (EIT)

A normally opaque medium becomes transparent due to the formation of a dark state.

6.3 STIRAP (Stimulated Raman Adiabatic Passage)

Efficient population transfer between quantum states without populating the excited state.

6.4 Quantum Information

Used in decoherence-free subspaces and quantum memory.

7 Conclusion

Dark states arise from quantum interference and represent a fundamental concept in controlling light-matter interactions. Their ability to suppress transitions makes them powerful tools in modern quantum technologies.